

RAK19018 WisBlock PoE Module for RAK13800 Datasheet

Overview

Description

The RAK19018 is a Power-over-Ethernet module used together with the RAK13800 Ethernet Interface module to draw power from CAT5/CAT6 cables. This PoE module is based on the Silvertel Ag9905MT converter board and compatible with the IEEE 802.3af PoE standard.

The Ag9905MT's signature and control circuit provide the PoE compatibility requirement by the Power Sourcing Equipment (PSE) before applying up to 9 Watts of power to the port. This provides a Class 0 signature.

The DC/DC converter operates over a wide input voltage from 36 V to 57 V and provides a regulated 5 V output with built-in short-circuit output protection.

Features

- Based on Silvertel Ag9905MT
- Requires RAK13800 Ethernet Module to work
- Power-over-Ethernet Module
- IEEE802.3af compliant
- Input voltage range : 36 V to 57 V
- Output voltage : 5 V
- Output power: 9 W
- Short-circuit protection and Over temperature protection
- Size: 32 mm x 25 mm

Specifications

Overview

Board Overview

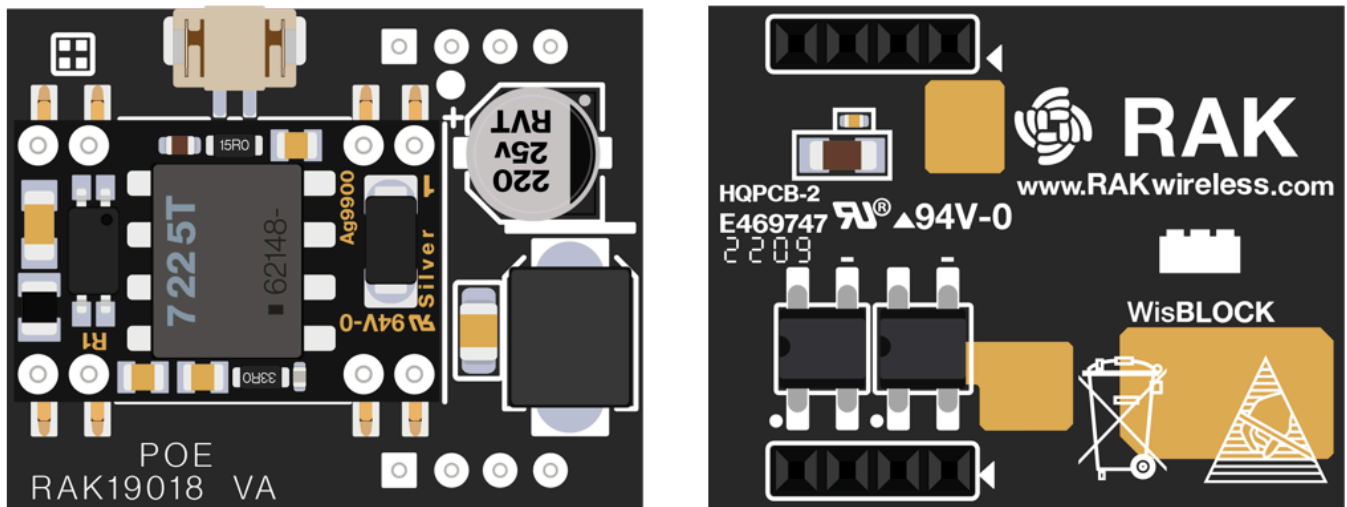


Figure 1: RAK19018 WisBlock PoE Module top (left) and bottom (right) view

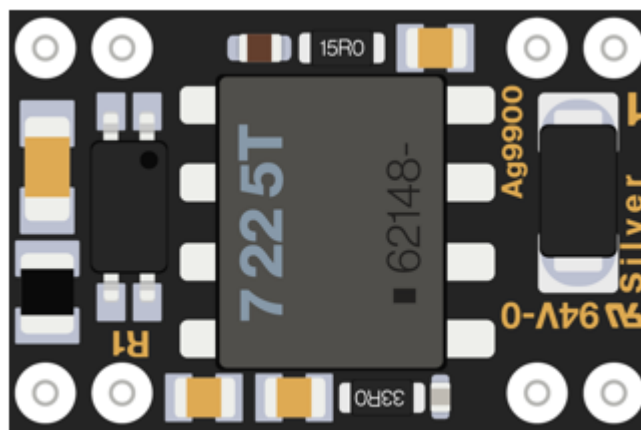


Figure 2: Ag9905MT PoE converter board

Mounting

The RAK19018 module is mounted on top of WisBlock Ethernet module RAK13800. These modules are attached via compatible header pins.

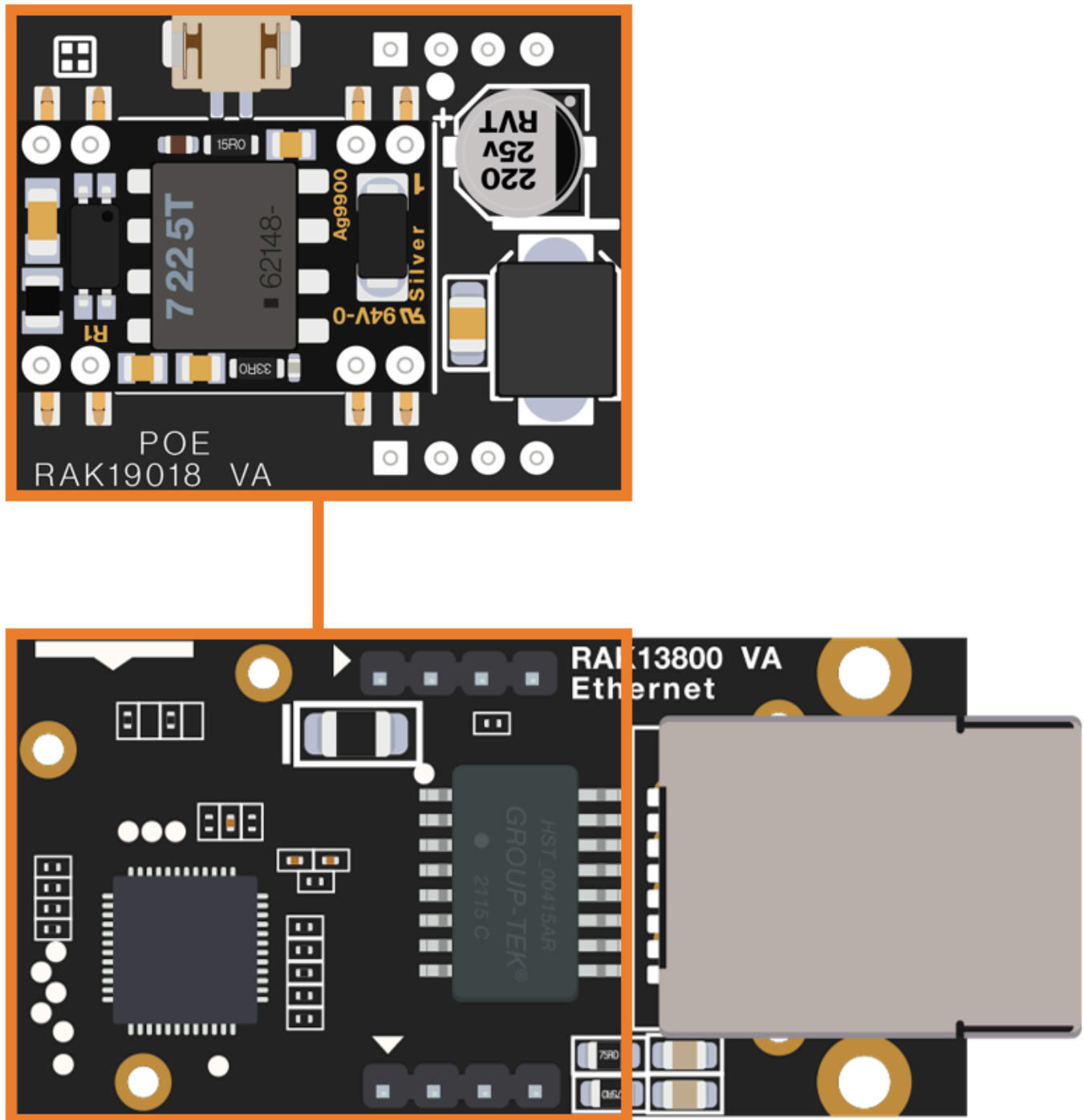


Figure 3: RAK19018 connection to RAK13800

Hardware

The hardware specification is categorized into five (5) parts. It shows the chipset, pinouts, and diagram of the module. It also presents the electrical and mechanical characteristics of the RAK19018 WisBlock PoE Module.

Chipset

Vendor	Part number
Silvertel	Ag9905MT

Pin Definition

The RAK19018 PoE Module has two connectors for 5 V output - **J3** and **J6**. The PoE input voltage is fed via **J5**.

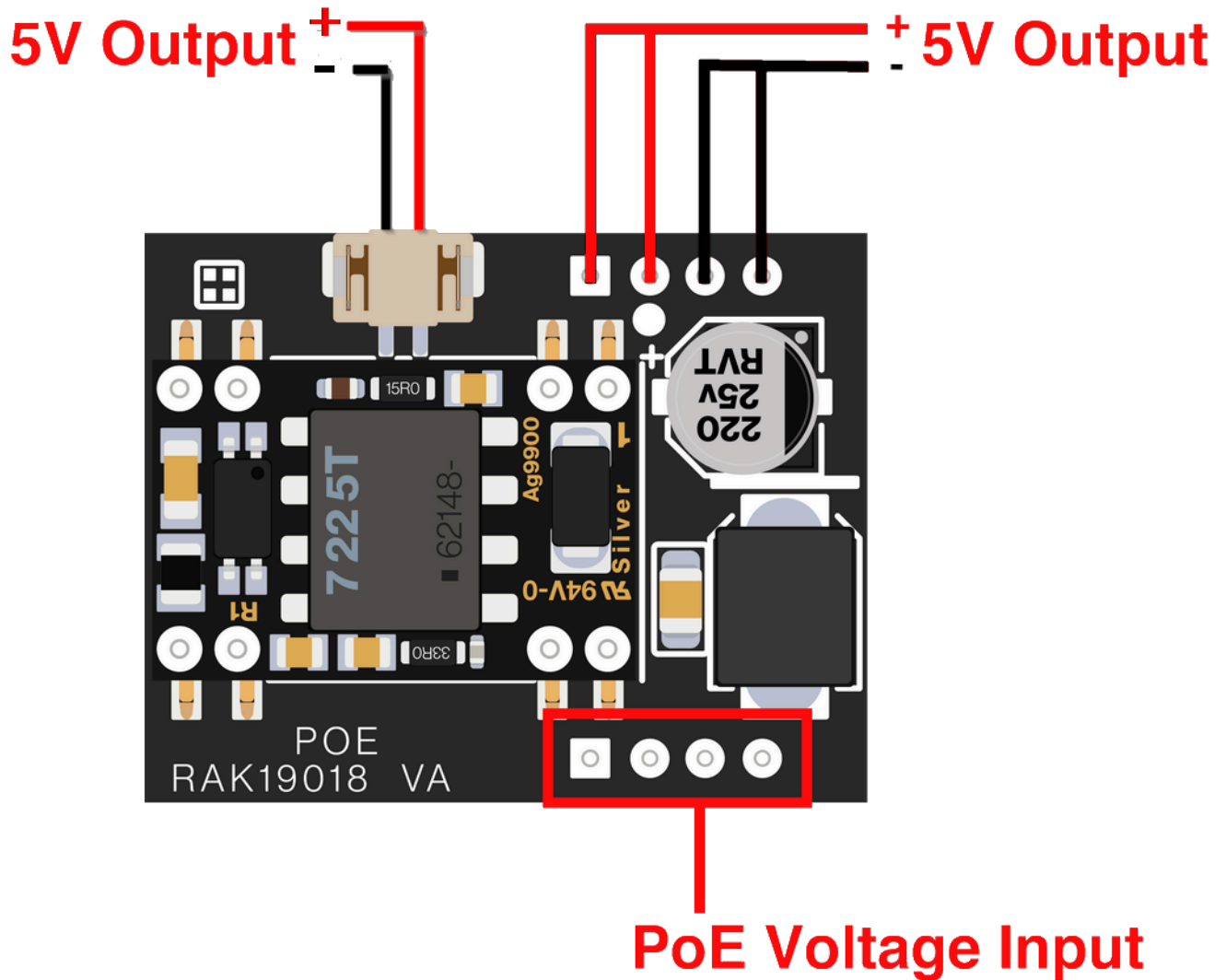


Figure 4: RAK19018 WisBlock Module pinout

Electrical Characteristics

This section shows the maximum and minimum ratings of the RAK19018 module and its recommended operating conditions.

Recommended Operating Conditions

Parameter	Min.	Nom.	Max.	Unit
Input Supply Voltage	36	48	57	V
Under Voltage Lockout	30		36	V
Nominal Output Voltage	4.75	5.0	5.25	V
Minimum Load	200			mA
Output Current (VIN = 48V)			1.8	A

Mechanical Characteristics

Board Dimensions

The mechanical dimension of RAK19018 is shown in **Figure 5**.

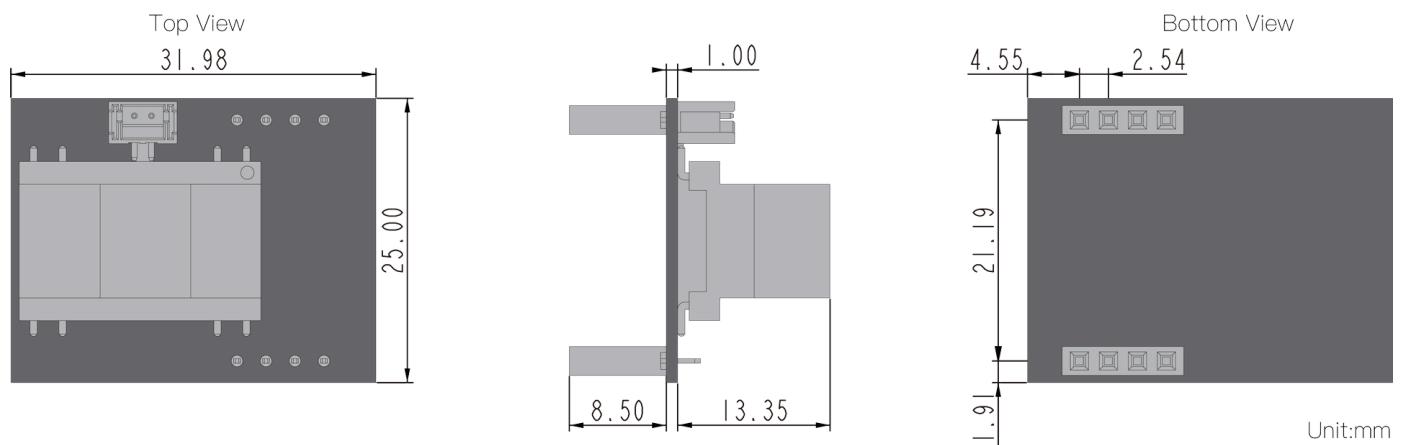


Figure 5: RAK19018 Mechanical Dimensions

Schematic Diagram

Figure 6 shows the schematic of the RAK19018 PoE module.

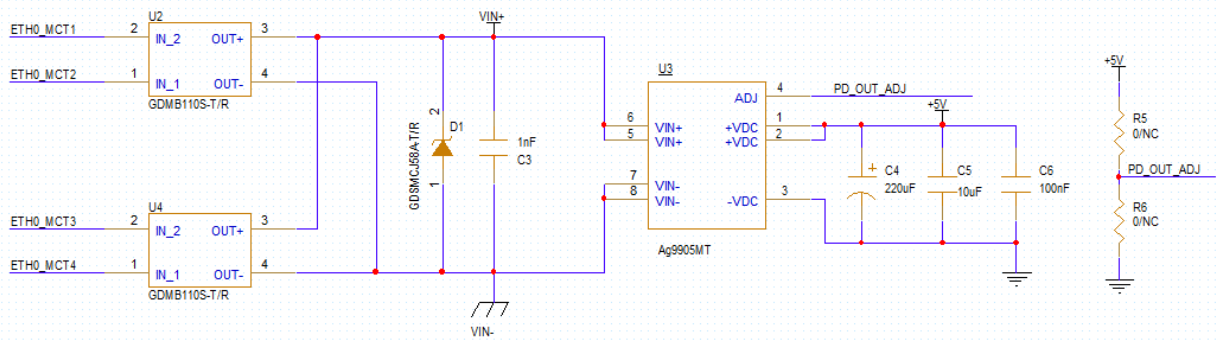


Figure 6: PoE Module

Using a cable, RAK19018 can power WisBlock Base boards like RAK5005-0.

- **J3** is a connector for 5 V output.
- **J5** and **J6** are used for connecting to RAK13800.

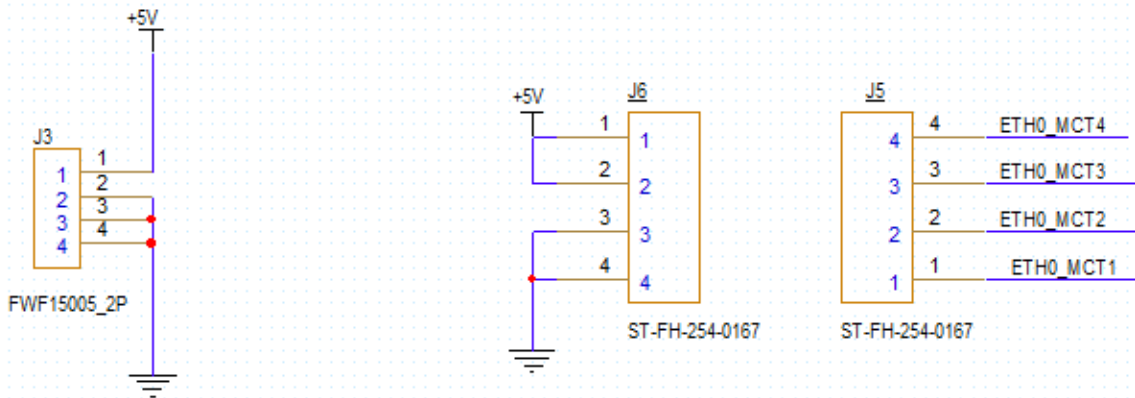


Figure 7: Connectors